Item 16: size-gated equivariant oracle

Verdict

ADVANCES, level constant, barrel-oracle level, toy sizes (n in {5, 7, 11, 13}). The frozen production cells n in {41, 47, 53, 61} are BLOCKED, not measured. The ADVANCES line must not be quoted as the answer to the frozen question, and it is not a production-oracle result.

Frozen rule (`PROTOCOL.md`) applied by `examples/blindspots_16_gate.rs` to the 32 held-out rows with `N0 = 61` tuned on the 48 training rows only:

check	result
(i) all gates (I, C, I2 root sets vs brute force)	pass, 0 failures in 48 training and 32 held-out rows (F control also 0)
(ii) training positive controls	loss seen at n=5 (I/C 0.2045, CI [0.1632, 0.2539]); win seen at n=13 (I/C 10.08, CI [9.66, 10.53])
(iii) pooled gated/incumbent CPU, CI lower bound > 1	2.4465, CI [2.3950, 2.5009]
(iv) no size regresses	no size has a CI upper bound < 1 (n=5, 7 are exactly 1 by construction)
(v) counted unit, pooled CI lower > 1 at w = 0, 0.1, 1	1.9586 [1.9472, 1.9711], 1.9584 [1.9470, 1.9710], 1.9570 [1.9456, 1.9695]: holds at all three

Gap to matched rho: not applicable. This is a within-method comparison of one repository solver on identical targets; no GAE row and no rho row exists (PROTOCOL.md, “Class”), so there is no distance-remaining figure to report and section 5a pricing ranks no arm. The three-price band is applied only to the GF(2^d) products inside the counted unit (above).

What the number means (read before quoting)

- **The gate does one thing.** `N0 = 61` switches the equivariant pass off at n = 5, 7 (31, 41 unknowns) and on at n = 11, 13 (61, 71). Below N0 the gated arm is the incumbent’s code, so its ratio is 1 by construction (Amendment A1). All of the gain is the gated-on sizes, where the gated arm is arm C. The pooled 2.45x is a geometric mean over 16 targets at ratio 1 and 16 targets at 3.4 to 10.4, so it is diluted by design; the ratio of summed CPU is 8.12x [7.72, 8.57], dominated by n = 13.
- **Most of the win at n >= 11 is fall-row augmentation, not equivariance.** The attribution arm F (the incumbent’s own degree-3 fall rows, no Frobenius symmetry) has held-out F/C of 1.135 [1.081, 1.193] at n = 11 and 1.342 [1.262, 1.430] at n = 13. Derived (point values, no interval): incumbent/F = (a/c)/(F/C) = 3.03 at n = 11 and 7.76 at n = 13. So of the 3.44x and 10.40x, about 3.0x and 7.8x comes from fall rows and only 1.13x and 1.34x from the equivariant structure itself. Item 02 measured F regressing 0.29 to 0.59x on the pipeline’s own systems, so the mechanism that carries the gain is not known to transfer to production.
- **Counted unit agrees in direction but not in size.** Counted a/c is 2.14x at n = 11 and 6.84x at n = 13 (w = 1), below the CPU ratios 3.44x and 10.40x. The solver’s word XORs are 2.1x to 6.9x lower; the rest of the CPU gain is not in the unit (barrel build and orbit enumeration are charged by CPU only). The w = 0 / 0.1 / 1 band moves the counted ratio by under 0.3 percent, so the verdict does not depend on the unpriced GF(2^d) products.
- **N0 sits on a boundary of the tested sizes.** Training J(T): 0.078 (T=31), 1.665 (41), 3.557 (61), 2.311 (71), 0 (never). The optimum is 61 because C wins at n = 11 and loses at n = 7; any N0 in 42..61 gives the same decisions on these four sizes. Four sizes cannot locate a crossover more finely, and no claim about other sizes is made.

Method

Instrument `examples/blindspots_16_size_gated_oracle.rs` (item 02’s barrel system, equivariant degree-3 pass, fall rows and incumbent oracle, rebuilt around the repository’s solver modules by `#[path]`, unmodified). Arms per target, in one process, random order, single thread, `prlimit --as=4 GiB`: I (incumbent), C (equivariant pass + solver), F (fall-row control), I2 (A/A). The gated arm is derived from the measured I and C. Metric: thread CPU seconds including system build and assembly; secondary counted `F2` word XORs plus w x GF products. Ratios are per-target paired, geometric mean with 95 percent percentile bootstrap (10,000 resamples within size; pooled resamples targets within size). Training t = 0..11 (48 targets, cards 01 and 02), held-out t = 12..19 (32 targets, card 03). Cards CARD-BLINDSPOT-16-01/02/03, receipts in `raw/`, relays checked with `tools/task_card.py check-relay` (all three verbatim, no findings).

Numbers (held-out, 8 targets per size, CPU, geometric mean [95% CI])

n	barrel unknowns	arm used	gated/incumbent	always-eq/ incumbent	A/A (I/I2)	F/C
5	31	I	1 (exact)	0.2161 [0.1536, 0.2927]	0.9821 [0.9483, 1.0184]	0.7219 [0.6382, 0.7919]
7	41	I	1 (exact)	0.1585 [0.1169, 0.2052]	1.0565 [0.9829, 1.1822]	0.4646 [0.4009, 0.5260]
11	61	C	3.4434 [3.2596, 3.6396]	3.4434 [3.2624, 3.6376]	0.9744 [0.9035, 1.0572]	1.1348 [1.0813, 1.1930]
13	71	C	10.4039 [9.7757, 11.1576]	10.4039 [9.7663, 11.1432]	0.9981 [0.9599, 1.0399]	1.3415 [1.2620, 1.4296]

Counted-unit gated/incumbent at w = 1: n = 11 2.1444 [2.0991, 2.1968], n = 13 6.8399 [6.7763, 6.9133]; w = 0 and 0.1 differ in the third decimal (`raw/analysis-g16.jsonl`). Training I/C: 0.2045, 0.1509, 3.4757, 10.0826 at n = 5, 7, 11, 13. Item 02 reported 0.250, 0.168, 5.10, 15.5 under the older solver; the incumbent got cheaper (`d72e8b04d`), so the n = 11 and 13 gains shrank (5.10 to 3.48, 15.5 to 10.1).

Feasibility at the frozen cells (`raw/feas.jsonl`, arithmetic only): barrel unknowns 170 to 250 against the solver's 64-variable u64 monomials, 211 to 311 symmetric unknowns against the pass's 192-bit masks, dense degree-3 matrix lower bound 37.3, 82.4, 165.9, 377.3 GB against 4 GB.

Controls

- **A/A:** all four A/A CIs contain 1 (n = 7 upper end 1.18, a 5.7 percent point offset with a wide CI); no order or contention bias flagged. Training n = 11 A/A is 0.92 with CI [0.85, 1.005], touching 1, so up to 8 percent order noise is possible at that size; the n = 11 gain of 3.4x is far outside it.
- **Positive control:** both held on training rows (above).
- **Negative control (wrong gate):** `--force-n0 31` (equivariant always) gives TRADES, pooled 1.0526 [0.9384, 1.1719], regression at n = 5, 7; `--force-n0 1000` (never) gives MATCHES, pooled exactly 1. Both as designed.
- **Replication:** held-out rows `t = 12..15` at n = 5, 7, 11 and `t = 12` at n = 13 were seen in the smoke test (disclosed in PROTOCOL.md); nothing was tuned on them.

Deviations and caveats

- The frozen cells n in {41, 47, 53, 61} on production IC are BLOCKED (four reasons in PROTOCOL.md). The measured run is a declared substitute: barrel gauge-fixed oracle, toy sizes, a different object from the design's. No exponent statement (four sizes, two gated off).
- N0 is tuned on training targets of the same sizes it is tested on (Deviation 2): a target-level hold-out, weaker than a size-level one.
- Amendment A1 (gated cost below N0 is I's own measurement) changed before any card ran.
- The equivariant pass applies only to Frobenius-symmetric systems; the production pipeline never produces one (item 02), so the gate is dead code there. This result is not evidence of any improvement to the production index-calculus baseline or to the gap to rho.
- Host L0: shared 4-vCPU cloud container, no isolation, no PMU. CPU time is thread CPU time, not wall; A/A shows no bias at this resolution. Peak RSS 0.105 GB.
- The smoke binary and rows were not committed. `crypto_lib` does not compile at the branch base, so the build route is one `rustc` per card against pinned rlibs.
- The relay files `raw/relay-1.txt` and `raw/relay-3.txt` were assembled by the assess stage: header and command lines retyped from the executor's returned text and the tail rows copied from the committed `raw/*.jsonl` (the rows the executor echoed, bound by stdout sha256 in the receipts). `relay-2.txt` likewise. They passed `check-relay`; they are not byte copies of the executor's message.

Files

`PROTOCOL.md`, `card-0{1,2,3}.yaml`, `raw/` (rows, receipts, relays, `analysis-g16.jsonl`), `graph.svg`, `graph.png`, `plot.py` (layout only, reads `raw/analysis-g16.jsonl`), `REPORT.pdf`. Reproduce analysis: see PROTOCOL.md "Reproduce".

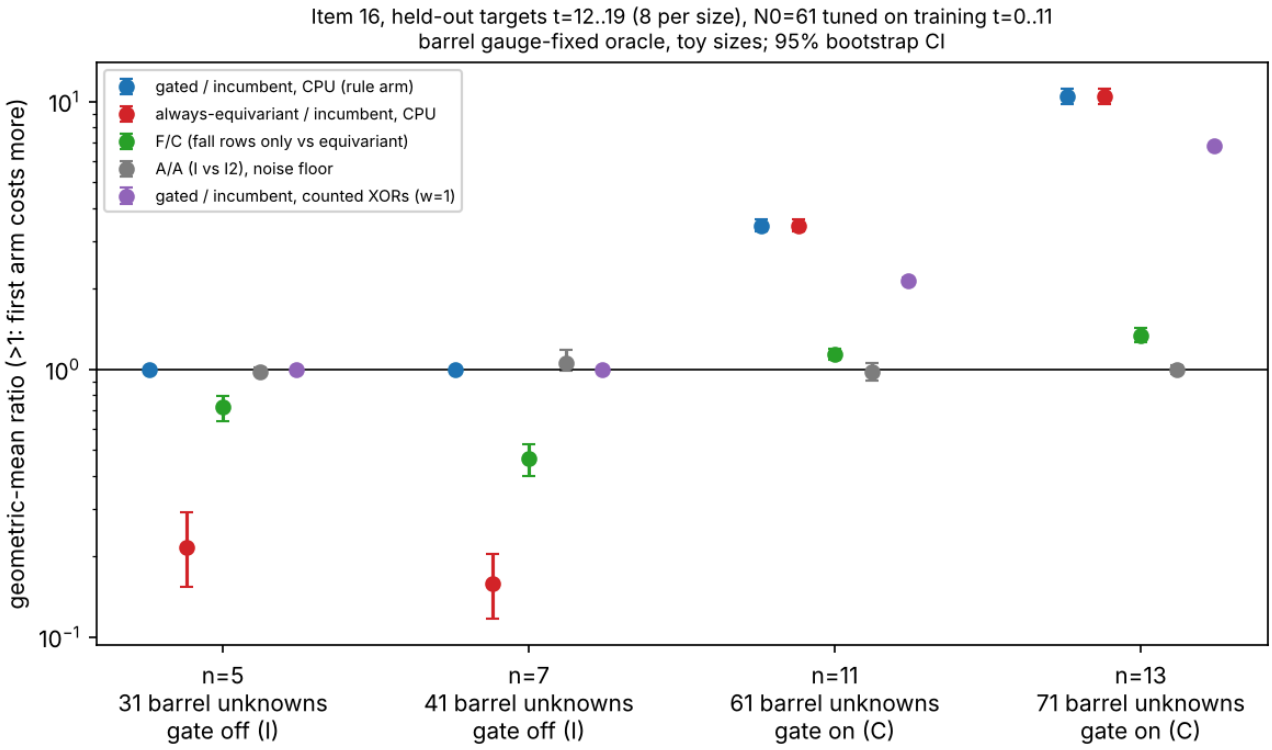


Figure 1: held-out ratios